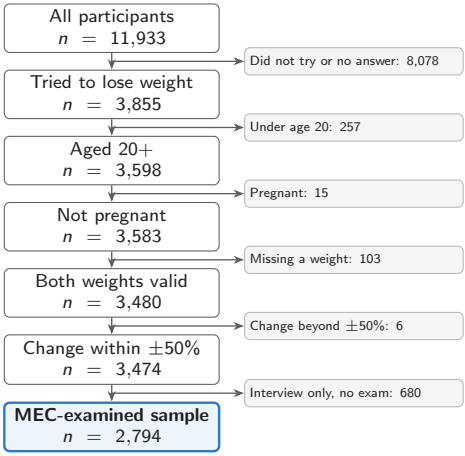
Data cleaning: 11,933 NHANES participants → 2,794 in the analysis



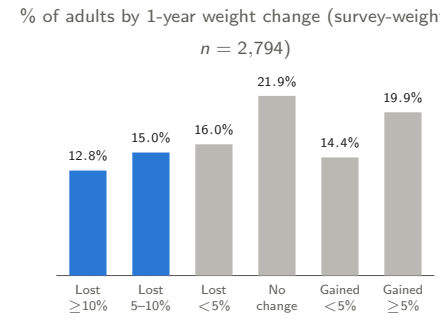
- **Merged 9 files** on `SEQN`: demographics, weight history, activity, diet, sleep, smoking, alcohol, PHQ-9, body measures (117 columns).
- **442 refused/don't-know codes → missing**, matched to answer length (7/9, 77/99, 777/999, 7777/9999); real 7s and 9s (e.g. 7 h sleep) kept.
- **Skip patterns filled, not imputed:** "none" → 0 or *Never* (vigorous activity: 51.5% → 0.5% missing).
- **Derived:** activity min/week, sedentary h/day, PHQ-9 score, drinks/week, smoking status, sleep h/night, BMI 1 year ago.
- **Dropped:** child-only, admin and redundant columns, and post-change body size (outcome leakage).

Outcome: % weight change
$= (W_{\text{now}} - W_{1\text{ yr ago}})/W_{1\text{ yr ago}} \times 100$
Features: 15 covariates + 12 behaviors

≈12% missing PHQ-9, drinks/week, income → multiple imputation; 2,117 complete cases (76%).

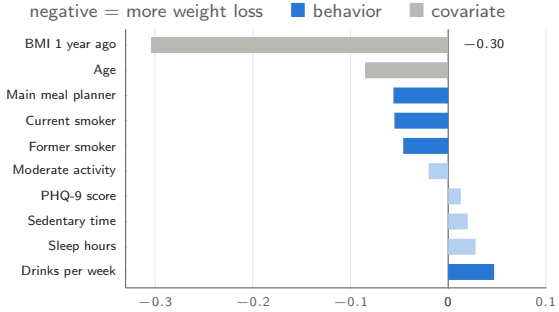
Exploration: starting BMI dominates; behaviors are weak on their own

28% lost ≥5%; 22% reported no change



Weighted mean −0.9%, median 0%; 34% gained. The spike at 0% is heaping (same round number reported twice).

Correlation with % weight change (Spearman r)



Faded = $p \geq 0.05$. No behavior exceeds $|r| = 0.06$; PHQ-9 and sleep look non-linear.

Next: nested models (covariates vs. + behaviors) · LASSO + gradient boosting with SHAP · lost ≥5% as a second outcome · MEC exam weights